Impact of Beyond-Diagonal RIS Inter-Panel Spacing in mmWave Communications

Hesam Nezhadmohamamd*, Reza Asvadi* , Abbas Pirhadi*
*Department of Telecommunications, Faculty of Electrical Engineering, Shahid Beheshti University, Tehran 1983963113, Iran
Email: h.nezhadmohammad@mail.sbu.ac. ir, {r_asvadi, a_pirhadi}@sbu.ac. ir

***Abstract*—Reconfigurable intelligent surface (RIS) is a promising technology for next-generation wireless networks. Recently, beyond-diagonal RIS (BD-RIS) has been proposed as a more general RIS architecture that enables interconnections between elements, which offers enhanced performance compared to conventional RIS. Despite recent advancements, the practical deployment of BD-RIS in millimeter-wave (mmWave) systems faces challenges that remain largely unexplored. To address this gap, we consider a multiuser outdoor-to-indoor mmWave downlink system with multiple BD-RIS panels and formulate a weighted sum-rate (WSR) maximization problem that treats the inter-panel spacing as a design variable. We propose a two-step solution combining a multi-panel extension of the partially proximal alternating direction method of multipliers algorithm with a derivative-free simplex search that iteratively updates the beamforming and BD-RIS susceptance matrices and adjusts the spacing. Simulation results reveal an optimal spacing that balances reduced channel correlation and path loss, and demonstrate that exploiting this optimum yields 20–25% higher WSR than a fixed-spacing baseline.**

I. Introduction

Ensuring high-speed wireless service to indoor users remains a persistent challenge for next-generation networks. While conventional solutions like indoor small-cell base stations are costly and complex, there is a growing interest in alternative strategies that leverage outdoor infrastructure. Outdoor-to-indoor millimeter-wave (mmWave) communication offers a promising path, benefiting from ample bandwidth for multi-gigabit data rates. However, mmWave signals face significant propagation hurdles, including high free-space path loss and limited penetration through building materials. Losses of 28–40 dB through walls and glass substantially restrict the reach of outdoor base stations to indoor environments [1].

Reconfigurable intelligent surface (RIS) offers a passive way to shape wireless propagation by reflecting signals toward desired directions [2]. Conventional RIS, which is based on a single-connected architecture, imposes a diagonal constraint on the scattering matrix, allowing each element to adjust only the phase of incident waves. While this creates virtual line-of-sight (LoS) paths, it restricts amplitude control and limits potential performance gains. Beyond-diagonal RIS (BD-RIS) architectures relax this constraint by introducing inter-element connections, yielding scattering matrices that are no longer diagonal [3], [4]. This structural generalization enables simultaneous control of phase and amplitude. BD-RIS can be implemented in various architectures offering design flexibility to balance hardware complexity with performance. Furthermore, they can operate in reflective, transmissive, or hybrid modes [4] and can also be deployed in a multi-sector setup to improve signal coverage [5].

Several recent works have explored the use of conventional RIS to enhance mmWave signal transmission into indoor environments, yet key limitations remain. Intelligent omni-surfaces mounted on building exteriors were optimized in [6] to improve energy efficiency, while [7] and [8] focused on positioning and orienting refracting or reflecting RIS panels to expand indoor coverage. However, these studies rely on diagonal phase-shift models and therefore cannot capture the richer interconnections between elements enabled by BD-RIS architectures. In addition, the spacing between multiple panels, a parameter that directly influences both channel correlation and overall path loss, is typically dictated by building geometry and not treated as a design variable. As a result, the potential advantages of considering panel spacing together with BD-RIS architectures remain largely unexplored.

This work addresses these gaps by evaluating BD-RIS performance under realistic conditions in a multiuser multi-input single-output (MU-MISO) mmWave system with multiple BD-RIS panels on an exterior wall. To enable joint optimization across panels, we derive a multi-panel BD-RIS system that concatenates all panels into an equivalent aggregate BD-RIS representation by preserving the sparsity patterns of different BD-RIS architectures. In addition, the proposed algorithm extends the existing partially proximal alternating direction method of multipliers (pp-ADMM) algorithm [9] to support multiple BD-RIS panels and explicitly optimize inter-panel spacing rather than assuming it to be a fixed geometric parameter, thereby providing a more complete framework for spatially aware BD-RIS optimization. Based on this model, we formulate a weighted sum-rate (WSR) maximization problem under transmit power, BD-RIS scattering matrix, and inter-panel spacing constraints. To solve it efficiently, we propose a two-step algorithm. For a fixed inter-panel spacing, the BS beamforming and BD-RIS scattering matrices are jointly optimized via the extended pp-ADMM algorithm, while the spacing itself is updated using a derivative-free method. At each iteration, the pp-ADMM subroutine evaluates candidate spacings and guides the simplex search by reflection, expansion, or contraction until convergence [10].

Our study reveals a key trade-off; increasing the spac-

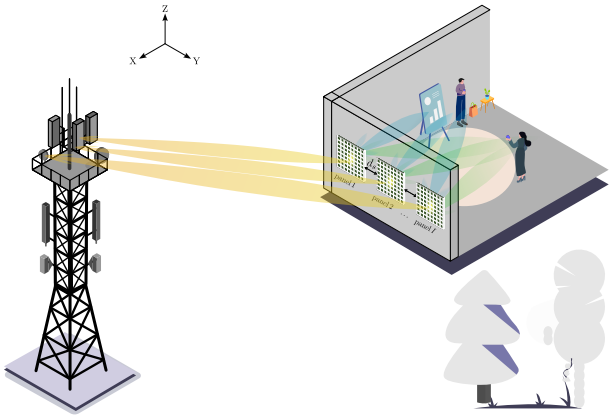


Fig. 1: An outdoor-to-indoor mmWave communication system with multiple BD-RIS panels separated by d_s.

ing between panels reduces correlation among the effective channels, thereby improving the WSR, yet simultaneously increases the propagation distance and thus the path loss. We demonstrate that the WSR initially improves with spacing, reaching a peak at a critical separation point. Beyond this point, further increases in spacing lead to detrimental path loss and performance degradation. By evaluating stem-connected, tree-connected, and single-connected BD-RIS architectures across different numbers of panels and users, we show that our proposed two-step optimization method effectively finds this critical inter-panel spacing to achieve near-optimal WSR. To the best of our knowledge, this work is the first to investigate the impact of inter-panel spacing on BD-RIS performance in a multi-panel BD-RIS-assisted outdoor-to-indoor mmWave system.

II. System Model

We consider a downlink MU-MISO outdoor-to-indoor mmWave system comprising a BS equipped with N transmit antennas, K single-antenna indoor users, and I BD-RIS panels deployed along the exterior building wall. Panel i contains M_i controllable elements and is modelled by a tunable susceptance matrix $\mathbf{B}_i \in \mathbb{R}^{M_i \times M_i}$ that characterises the impedance network among its elements [11]. The susceptance matrices $\{\mathbf{B}_i\}_{i=1}^{I}$ are symmetric real-valued matrices belonging to an architecture-specific set $\mathcal{B}$:

$$\mathcal{B} = \{\mathbf{B}_i \in \mathbb{R}^{M \times M} \mid \mathbf{B}_i = \mathbf{B}_i^T,\ [\mathbf{B}_i]_{n,\mathcal{S}_n} = \mathbf{0}\},$$

where $\mathcal{S}_n$ denotes the set of indices not connected to element n, i.e., $m \in \mathcal{S}_n$ if and only if there is no interconnection between the n-th and m-th elements [9].

A. Channel Model

We assume that the direct BS–user link is blocked by the exterior wall and is therefore neglected. For panel i, let $\mathbf{G}_i \in \mathbb{C}^{M_i \times N}$ denote the BS–RIS channel and let $\mathbf{h}_{i,k} \in \mathbb{C}^{M_i}$ denote the RIS–user k channel. Following [12], both links are modeled by an SV geometric mmWave channel with a finite number of resolvable paths, where $\ell = 1$ corresponds to the LoS path and $\ell \geq 2$ correspond to NLoS paths. The channel from the BS to panel i is given by

$$\begin{aligned}\mathbf{G}_i = \sqrt{\frac{NM_i}{L_{\mathrm{BR}}}}\Big(&\beta_{i,1}\,\mathbf{a}_{\mathrm{RIS},i}(\gamma^r_{i,1},\eta^r_{i,1})\,\mathbf{a}^H_{\mathrm{BS}}(\phi^t_{i,1}) \\ &+ \sum_{\ell=2}^{L_{\mathrm{BR}}} \beta_{i,\ell}\,\mathbf{a}_{\mathrm{RIS},i}(\gamma^r_{i,\ell},\eta^r_{i,\ell})\,\mathbf{a}^H_{\mathrm{BS}}(\phi^t_{i,\ell})\Big),\end{aligned} \tag{1}$$

where L_{BR} is the number of resolvable paths for the BS–RIS link, and $\beta_{i,\ell}$ is the complex gain of path ℓ. The complex path gains satisfy

$$\beta_{i,1} \sim \mathcal{CN}\Big(0, 10^{-0.1\kappa_i^{\mathrm{LOS}}}\Big), \tag{2a}$$

$$\beta_{i,\ell} \sim \mathcal{CN}\Big(0, 10^{-0.1\kappa_i^{\mathrm{NLOS}}}\Big), \quad \ell = 2,\ldots,L_{\mathrm{BR}}, \tag{2b}$$

with path losses

$$\kappa_i^{\mathrm{LOS}} = a_{\mathrm{LOS}} + 10b_{\mathrm{LOS}}\log_{10}(d_{\mathrm{BR}_i}) + \mu_{\mathrm{LOS}}, \tag{3a}$$

$$\kappa_i^{\mathrm{NLOS}} = a_{\mathrm{NLOS}} + 10b_{\mathrm{NLOS}}\log_{10}(d_{\mathrm{BR}_i}) + \mu_{\mathrm{NLOS}}, \tag{3b}$$

where d_{BR_i} is the BS–panel distance, and $\mu_{\mathrm{LOS}} \sim \mathcal{N}(0,\sigma^2_{\mathrm{LOS}})$ and $\mu_{\mathrm{NLOS}} \sim \mathcal{N}(0,\sigma^2_{\mathrm{NLOS}})$ model log-normal shadow fading [12]. Similarly, the channel from panel i to user k is modeled as

$$\begin{aligned}\mathbf{h}_{i,k} = \sqrt{\frac{M_i}{L_{\mathrm{RU}}}}\Big(&\alpha_{i,k,1}\,\mathbf{a}_{\mathrm{RIS},i}(\gamma^r_{i,k,1},\eta^r_{i,k,1}) \\ &+ \sum_{\ell=2}^{L_{\mathrm{RU}}} \alpha_{i,k,\ell}\,\mathbf{a}_{\mathrm{RIS},i}(\gamma^r_{i,k,\ell},\eta^r_{i,k,\ell})\Big),\end{aligned} \tag{4}$$

where L_{RU} is the number of resolvable paths for the RIS–user link. The path gains $\alpha_{i,k,\ell}$ follow the same statistics as in (2)–(3) after replacing d_{BR_i} by the panel–user distance $d_{R_{i,k}}$. Assuming half-wavelength spacing, the BS uniform linear array (ULA) response is

$$\mathbf{a}_{\mathrm{BS}}(\phi) = \frac{1}{\sqrt{N}}\left[1, e^{j\pi\sin(\phi)},\ldots,e^{j\pi(N-1)\sin(\phi)}\right]^T. \tag{5}$$

For panel i with a $M_{x,i} \times M_{y,i}$ uniform rectangular array (URA), where $M_i = M_{x,i} \times M_{y,i}$, the element index is defined as $p = (n-1)M_{x,i} + m$, with $m = 1,\ldots,M_{x,i}$ and $n = 1,\ldots,M_{y,i}$. The URA response is

$$[\mathbf{a}_{\mathrm{RIS},i}(\gamma,\eta)]_p = \frac{1}{\sqrt{M_i}}\,e^{j\pi(m-1)\cos(\eta)\sin(\gamma)}\,e^{j\pi(n-1)\sin(\eta)}. \tag{6}$$

The LoS angles are determined by geometry, while the NLoS angles are drawn from the chosen angular distributions. For notational convenience, we collect the RIS–user channels of panel i into $\mathbf{H}_i = [\mathbf{h}_{i,1},\ldots,\mathbf{h}_{i,K}] \in \mathbb{C}^{M_i \times K}$. Each panel applies a complex scattering matrix

$$\mathbf{\Theta}_i = (\mathbf{I}_{M_i} + jZ_0\mathbf{B}_i)^{-1}(\mathbf{I}_{M_i} - jZ_0\mathbf{B}_i), \tag{7}$$

where $Z_0 = 50\,\Omega$ is the reference impedance and j denotes the imaginary unit. The composite channel from the BS to user k through panel i is then $\mathbf{h}^H_{i,k}\mathbf{\Theta}_i\mathbf{G}_i$.

In the system where we have multiple BD-RIS panels, the inter-panel spacing d_s impacts the channels seen

by the BS and the users by inducing path-loss and array response variations across panels. We thus define the aggregated channel matrices as follows. Let $\bar{\mathbf{G}}(d_s) = [\mathbf{G}_1, \ldots, \mathbf{G}_I]^T \in \mathbb{C}^{M_{\text{tot}} \times N}$, $\bar{\mathbf{H}}(d_s) = [\mathbf{H}_1, \ldots, \mathbf{H}_I]^T \in \mathbb{C}^{M_{\text{tot}} \times K}$, and the aggregated block-diagonal scattering matrix $\bar{\mathbf{\Theta}} = \text{blkdiag}(\mathbf{\Theta}_1, \ldots, \mathbf{\Theta}_I) \in \mathbb{C}^{M_{\text{tot}} \times M_{\text{tot}}}$, where $M_{\text{tot}} = \sum_{i=1}^{I} M_i$ denotes the total number of RIS elements across all panels. We further define the aggregated susceptance matrix $\bar{\mathbf{B}} = \text{blkdiag}(\mathbf{B}_1, \ldots, \mathbf{B}_I)$ and denote by $\bar{\mathcal{B}}$ the set of all block-diagonal matrices with blocks $\mathbf{B}_i \in \mathcal{B}$. For user k the aggregate RIS-user channel is the k-th column of $\bar{\mathbf{H}}(d_s)$, denoted $\bar{\mathbf{h}}_k(d_s) \in \mathbb{C}^{M_{\text{tot}}}$. Using the aggregated notation, the effective channel from the BS to user k is $\bar{\mathbf{h}}_k^H(d_s)\bar{\mathbf{\Theta}}\bar{\mathbf{G}}(d_s) \in \mathbb{C}^{1 \times N}$ which succinctly captures the superposition of the contributions from all panels.

B. Received Signal and Achievable Rate

Let $\mathbf{w}_k \in \mathbb{C}^N$ denote the downlink beamforming vector for user k and s_k the corresponding information symbol with $\mathbb{E}\{|s_k|^2\} = 1$. Neglecting direct links, the received signal at user k is obtained by summing the contributions through all panels:

$$\begin{aligned} y_k &= \sum_{i=1}^{I} \mathbf{h}_{i,k}^H \mathbf{\Theta}_i \mathbf{G}_i \mathbf{w}_k s_k \; + \sum_{j \neq k} \sum_{i=1}^{I} \mathbf{h}_{i,k}^H \mathbf{\Theta}_i \mathbf{G}_i \mathbf{w}_j s_j + n_k \\ &= \bar{\mathbf{h}}_k^H(d_s)\bar{\mathbf{\Theta}}\bar{\mathbf{G}}(d_s)\mathbf{w}_k s_k + \sum_{j \neq k} \bar{\mathbf{h}}_k^H(d_s)\bar{\mathbf{\Theta}}\bar{\mathbf{G}}(d_s)\mathbf{w}_j s_j + n_k, \end{aligned} \tag{8}$$

where $n_k \sim \mathcal{CN}(0, \sigma^2)$ is the complex Gaussian noise. The signal-to-interference-plus-noise ratio (SINR) of user k is

$$\gamma_k(\mathbf{\Theta}, \mathbf{W}, d_s) = \frac{\left|\bar{\mathbf{h}}_k^H(d_s)\bar{\mathbf{\Theta}}\bar{\mathbf{G}}(d_s)\mathbf{w}_k\right|^2}{\sum_{j \neq k}\left|\bar{\mathbf{h}}_k^H(d_s)\bar{\mathbf{\Theta}}\bar{\mathbf{G}}(d_s)\mathbf{w}_j\right|^2 + \sigma^2}, \tag{9}$$

Directly optimizing $\{\mathbf{B}_i\}_{i=1}^I$ and the beamforming matrix $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_K]$ in (9) is challenging because $\bar{\mathbf{\Theta}}$ depends on the susceptance matrices through the matrix inversion in (7). Following [9], [13], we introduce auxiliary variables that linearly relate the RIS-user channels to the susceptance matrices and thereby eliminate the inversion. For each user k we define

$$\bar{\mathbf{u}}_k(d_s) = \bar{\mathbf{\Theta}}^H \bar{\mathbf{h}}_k(d_s) \in \mathbb{C}^{M_{\text{tot}}}, \tag{10}$$

and collect them as $\bar{\mathbf{U}}(d_s) = [\bar{\mathbf{u}}_1(d_s), \ldots, \bar{\mathbf{u}}_K(d_s)] \in \mathbb{C}^{M_{\text{tot}} \times K}$. Using the relation in (7), one can show that the variables $\bar{\mathbf{U}}(d_s)$ and $\bar{\mathbf{H}}(d_s)$ satisfy a bilinear constraint

$$(\mathbf{I}_{M_{\text{tot}}} - jZ_0\bar{\mathbf{B}})\,\bar{\mathbf{U}}(d_s) = (\mathbf{I}_{M_{\text{tot}}} + jZ_0\bar{\mathbf{B}})\,\bar{\mathbf{H}}(d_s). \tag{11}$$

By substituting (10) and (11) into the SINR definition in (9), we obtain

$$\gamma_k(\mathbf{u}_k, \mathbf{W}, d_s) = \frac{\left|\bar{\mathbf{u}}_k^H(d_s)\bar{\mathbf{G}}(d_s)\mathbf{w}_k\right|^2}{\sum_{j \neq k}\left|\bar{\mathbf{u}}_k^H(d_s)\bar{\mathbf{G}}(d_s)\mathbf{w}_j\right|^2 + \sigma^2}, \tag{12}$$

and $R_k(\mathbf{u}_k, \mathbf{W}, d_s) = \log_2(1 + \gamma_k(\mathbf{u}_k, \mathbf{W}, d_s))$ denotes the achievable rate of user k.

III. WSR Maximization with Multiple BD-RIS Panels

The WSR maximization problem with multiple BD-RIS panels can now be formulated as

$$\max_{\mathbf{W}, \bar{\mathbf{B}}, \bar{\mathbf{U}}, d_s} \quad \sum_{k=1}^{K} \omega_k R_k(\mathbf{u}_k, \mathbf{W}, d_s) \tag{13a}$$

$$\text{s.t.} \quad (\mathbf{I}_{M_{\text{tot}}} - jZ_0\bar{\mathbf{B}})\,\bar{\mathbf{U}}(d_s) = (\mathbf{I}_{M_{\text{tot}}} + jZ_0\bar{\mathbf{B}})\,\bar{\mathbf{H}}(d_s), \tag{13b}$$

$$\bar{\mathbf{B}} \in \bar{\mathcal{B}}, \tag{13c}$$

$$\sum_{k=1}^{K} \|\mathbf{w}_k\|_2^2 \leq P_{\max}, \tag{13d}$$

$$d_s \in [d_{\min},\, d_{\max}], \tag{13e}$$

where ω_k is a non-negative weight indicating the priority of user k. The parameters $d_{\min}$ and $d_{\max}$ specify the lower and upper bounds on d_s, respectively. Directly solving (13) is challenging due to the coupling between the variables and the bilinear constraint (13b). We adopt a two-step strategy that alternates between updating the optimization variables for a fixed spacing via pp-ADMM algorithm and adjusting the spacing via a derivative-free simplex search method.

The WSR formulation (13) shares the same structure as the single-BD-RIS problem in [9] when the inter-panel spacing constraint (13e) is neglected. Consequently, the optimization method developed therein can be extended to multiple panels using the aggregated variables $\bar{\mathbf{B}}$, $\bar{\mathbf{G}}(d_s)$, $\bar{\mathbf{H}}(d_s)$, and $\bar{\mathbf{U}}(d_s)$; see [9] for the detailed implementation of the pp-ADMM algorithm.

To optimize the inter-panel spacing d_s, we employ the Nelder-Mead (NM) simplex search method [10], [14]. The objective depends on d_s through the channel vectors $\{\bar{\mathbf{u}}_k(d_s)\}_{k=1}^K$ and the channel matrix $\bar{\mathbf{G}}(d_s)$, and does not admit a tractable gradient with respect to d_s. The NM method is therefore suitable for this problem due to its geometrical sampling property. Let N_s be the number of candidate vertices in the simplex. The simplex thus contains $N_s + 1$ vertices that are randomly chosen within $[d_{\min}, d_{\max}]$. At iteration index $q = 0$, this constructs the initial spacing set $\mathcal{S}^{(0)} = \{d_{s,1}^{(0)}, \ldots, d_{s,n_s}^{(0)}, \ldots, d_{s,N_s+1}^{(0)}\}$, with $n_s \in \{1, 2, \ldots, N_s + 1\}$, where each element represents a random inter-panel spacing d_s.

At the q-th NM iteration, each candidate spacing $d_{s,n_s}^{(q)}$ is evaluated via the pp-ADMM algorithm, yielding the objective value $f(d_{s,n_s}^{(q)})$. The vertices are then sorted in ascending order by these objective values such that $f(d_{s,1}^{(q)}) \leq f(d_{s,2}^{(q)}) \leq \cdots \leq f(d_{s,N_s+1}^{(q)})$. The best point is the vertex with the lowest value, $d_{s,1}^{(q)}$, and the worst point is the vertex with the highest value, $d_{s,N_s+1}^{(q)}$[1]. The centroid of the best N_s points after excluding

[1]We remark that NM algorithm inherently performs minimization. To adapt it to our WSR maximization objective (see (13a)), we minimize the negative of the WSR.

the worst point is

$$d_{s,c}^{(q)} = \frac{1}{N_s} \sum_{n_s=1}^{N_s} d_{s,n_s}^{(q)}. \tag{14}$$

The worst vertex $d_{s,N_s+1}^{(q)}$ is then replaced according to the standard NM transformations as shown in Algorithm 1. According to universal choices for NM parameters, we have $\alpha = 1$, $\gamma = 2$, $\rho = 0.5$ and $\sigma = 0.5$ as the reflection, expansion, contraction and shrinkage coefficients, respectively. Defining

$$\mathcal{D}_s^{(q)} = \left\{ \left| d_{s,i}^{(q)} - d_{s,j}^{(q)} \right| \, \middle| \, i,j \in \{1,2,\ldots,N_s+1\}, i \neq j \right\}, \tag{15}$$

which is the pairwise distance set between the inter-panel spacings across all N_s+1 spacings in the q-th iteration. When $\max(\mathcal{D}_s^{(q)}) \leq \tau$ holds, all the N_s+1 spacings lie within a small sphere with diameter τ, making further improvement less probable. Then, the NM method has converged, and the optimized inter-panel spacing $d_{s,1}^{(q)}$ in $\mathcal{S}^{(q)}$ yields the maximum objective function value is derived. The proposed two-step optimization framework is summarized in Algorithm 1.

IV. Numerical Results

In this section, we present the numerical results to evaluate the performance of our proposed algorithm. Unless otherwise specified, the system parameters follow Table I. The I BD-RIS panels are deployed at a common height, uniformly spaced along the y-axis and arranged symmetrically around a reference RIS panel. We assume that panels have the same number of elements, i.e., $M_i = M$, $\forall i \in \{1,2,\ldots,I\}$ and operate in transmissive mode which allows incident waves to pass through them [4]. We also adopt the stem width from [15] for the stem-connected case.

To analyze the impact of inter-panel spacing d_s on the WSR, Fig. 2 presents results for $K = I = 4$ and three RIS sizes $M \in \{16, 32, 64\}$. The curves reveal a clear trade-off between channel conditioning and distance-dependent path-loss. When the panels are too close, the LoS links from neighboring panels are highly correlated and the spatial degree of freedom (DoF) of the composite channel is limited. Increasing d_s improves the conditioning of the cascaded channels and thus enhances the WSR. For instance, at $M = 16$ the WSR of the stem-connected BD-RIS increases from about 7 bps/Hz at $d_s = 1$ m to approximately 8 bps/Hz at the optimal spacing $d_s^\star \approx 3$ m. The tree-connected and single-connected architectures exhibit a similar trend with slightly lower peak values (around 7.9 bps/Hz and 7.75 bps/Hz, respectively). Beyond this critical spacing, the additional path-loss outweighs the correlation reduction and the WSR decreases.

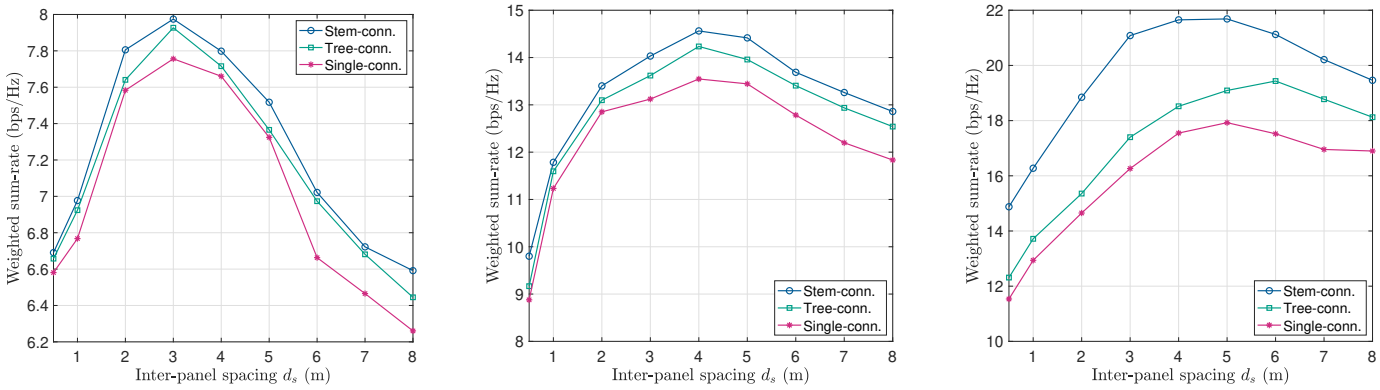
As the panel size grows from $M = 16$ to $M = 32$, both the achievable rate and the optimal spacing increase. For $M = 32$ the optimum shifts to $d_s^\star \approx$ 4–5 m and the stem-connected architecture attains a peak of about 14.56 bps/Hz, while the tree- and single-connected schemes achieve around 14.23 and 13.55 bps/Hz, respectively. When $M = 64$, the benefit of richer inter-element connectivity becomes more pronounced. The stem-connected architecture achieves a maximum WSR of roughly 21.69 bps/Hz at $d_s^\star \approx 5$ m, compared with 19.43 bps/Hz for the tree-connected and 17.93 bps/Hz for the single-connected architecture. These results emphasize that

Algorithm 1 Two-Step pp-ADMM-NM Algorithm for Spacing Optimization

Input: Initial simplex $\{d_s^{(n_s)}\}_{n_s=1}^{N_s+1}$ with $d_s^{(n_s)} \in [d_{\min}, d_{\max}]$, pp-ADMM tolerance ε, NM coefficients $(\alpha, \gamma, \rho, \sigma)$, simplex tolerance τ.

Set iteration counter $q = 0$.
repeat
  Step 1 - pp-ADMM updates:
  for $n_s = 1$ to $N_s + 1$ **do**
    Set spacing $d_s = d_s^{(n_s)}$ and initialize pp-ADMM variables $\bar{\mathbf{B}}^0$, $\bar{\mathbf{U}}^0$, $\mathbf{W}^0$, $\mathbf{y}^0$ and $\boldsymbol{\gamma}^0$ [9, Section III].
    Iterate the pp-ADMM updates until convergence tolerance ε is reached.
    Compute the objective value $f(d_s^{(n_s)})$ using (13a) with the obtained $\mathbf{W}$ and $\bar{\mathbf{U}}$.
  end for
  Step 2 - NM update:
  Sort the vertices so that $f(d_s^{(1)}) \leq f(d_s^{(2)}) \leq \cdots \leq f(d_s^{(N_s+1)})$.
  Compute the centroid $d_{s,c}^{(q)}$ of the best N_s points using (14).
  Reflection: $d_{s,r}^{(q)} = d_{s,c}^{(q)} + \alpha\,(d_{s,c}^{(q)} - d_{s,N_s+1}^{(q)})$; evaluate $f(d_{s,r}^{(q)})$ via a pp-ADMM run.
  if $f(d_{s,1}^{(q)}) \leq f(d_{s,r}^{(q)}) < f(d_{s,N_s}^{(q)})$ **then**
    Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,r}^{(q)}$.
  else if $f(d_{s,r}^{(q)}) < f(d_{s,1}^{(q)})$ **then**
    Expansion: $d_{s,e}^{(q)} = d_{s,c}^{(q)} + \gamma\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,e}^{(q)})$.
    if $f(d_{s,e}^{(q)}) < f(d_{s,r}^{(q)})$ **then**
      $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,e}^{(q)}$.
    else
      $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,r}^{(q)}$.
    end if
  else if $f(d_{s,N_s}^{(q)}) \leq f(d_{s,r}^{(q)}) < f(d_{s,N_s+1}^{(q)})$ **then**
    Outside contraction: $d_{s,\mathrm{oc}}^{(q)} \leftarrow d_{s,c}^{(q)} + \rho\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,\mathrm{oc}}^{(q)})$.
    if $f(d_{s,\mathrm{oc}}^{(q)}) \leq f(d_{s,r}^{(q)})$ **then**
      Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,\mathrm{oc}}^{(q)}$.
    else
      Shrinkage: For $i = 2,\ldots,N_s+1$ set $d_{s,i}^{(q)} \leftarrow d_{s,1}^{(q)} + \sigma\,(d_{s,i}^{(q)} - d_{s,1}^{(q)})$ and re-evaluate $f(d_{s,i}^{(q)})$ via pp-ADMM.
    end if
  else
    Inside contraction: $d_{s,\mathrm{ic}}^{(q)} \leftarrow d_{s,c}^{(q)} - \rho\,(d_{s,r}^{(q)} - d_{s,c}^{(q)})$; evaluate $f(d_{s,\mathrm{ic}}^{(q)})$.
    if $f(d_{s,\mathrm{ic}}^{(q)}) < f(d_{s,N_s+1}^{(q)})$ **then**
      Replace $d_{s,N_s+1}^{(q+1)} \leftarrow d_{s,\mathrm{ic}}^{(q)}$.
    else
      Shrinkage: For $i = 2,\ldots,N_s+1$ set $d_{s,i}^{(q)} \leftarrow d_{s,1}^{(q)} + \sigma\,(d_{s,i}^{(q)} - d_{s,1}^{(q)})$ and re-evaluate $f(d_{s,i}^{(q)})$ via pp-ADMM.
    end if
  end if
  $q \leftarrow q + 1$.
until $\max(\mathcal{D}_s^{(q)}) \leq \tau$

Output: Best spacing $d_s^\star = d_{s,1}^{(q)}$ and corresponding $\mathbf{W}^\star$, $\bar{\mathbf{B}}^\star$ and $\bar{\mathbf{U}}^\star$ obtained by pp-ADMM.



(a) $K = I = 4$, $M = 16$. (b) $K = I = 4$, $M = 32$. (c) $K = I = 4$, $M = 64$.

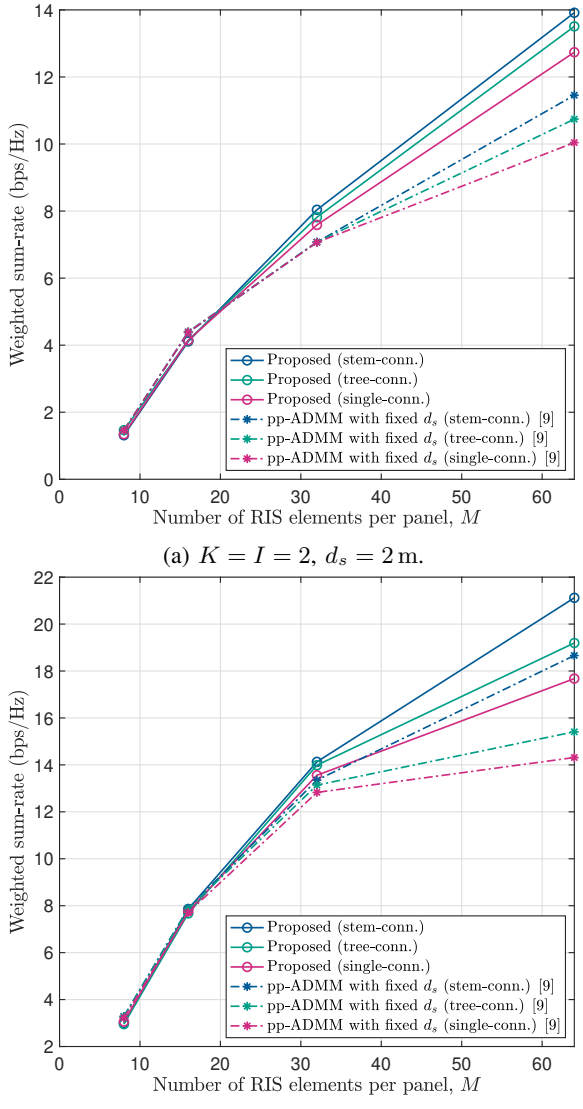
Fig. 2: WSR versus inter-panel spacing for stem-connected, tree-connected (tridiagonal), and single-connected BD-RIS.

there exists a non-trivial optimal spacing that depends on the surface size and architecture, and underline the importance of adaptively optimizing d_s rather than assuming it to be fixed.

Fig. 3 compares the WSR of the proposed two-step algorithm with that of the pp-ADMM baseline (which uses a fixed inter-panel spacing) for $K = I \in \{2, 4\}$ and $M \in \{8, 16, 32, 64\}$. In all cases the proposed method adaptively chooses the spacing within $[d_{\min}, d_{\max}]$ and therefore yields higher rates, especially for large panels. As shown in Fig. 3a for $K = I = 2$, the difference between the two approaches is modest at small M. Both algorithms achieve about 1.5 bps/Hz at $M = 8$ and 4.3 bps/Hz at $M = 16$ because the channels offer limited spatial DoF. However, as M increases the benefit of adaptive spacing becomes evident. At $M = 32$ the stem-connected BD-RIS with the proposed algorithm attains roughly 8.0 bps/Hz compared with about 7.0 bps/Hz for pp-ADMM. When $M = 64$ the gap widens further. Our method reaches about 13.92 bps/Hz for the stem-connected architecture, while the tree- and single-connected counterparts achieve around 13.51 and 12.74 bps/Hz, respectively. The pp-ADMM benchmarks, by contrast, yield roughly 11.45 bps/Hz (stem), 10.74 bps/Hz (tree) and 10.04 bps/Hz (single). Compared to the pp-ADMM benchmarks [9], our proposed algorithm achieves 21.58%, 25.80%, and 26.90% WSR improvement for stem-, tree-, and single-connected architectures, respectively.

For $K = I = 4$ (Fig. 3b), the WSR improvement stemming from the increased number of panels, and the resulting additional DoF, is evident[2]. At $M = 8$ and $M = 16$ the rates are about 3.0 bps/Hz and 7.7 bps/Hz, respectively, for both schemes. At $M = 32$ the proposed method yields approximately 14.13 bps/Hz, 13.99 bps/Hz and 13.56 bps/Hz for the stem-, tree- and single-connected architectures, respectively. The pp-ADMM baseline attains around 12.8–13.5 bps/Hz. At $M = 64$ the performance gaps become more significant with the proposed algorithm achieves roughly 21.12 bps/Hz for the stem-connected BD-RIS versus about 18.66 bps/Hz for pp-ADMM (a gain of 13.2%), while the tree- and single-

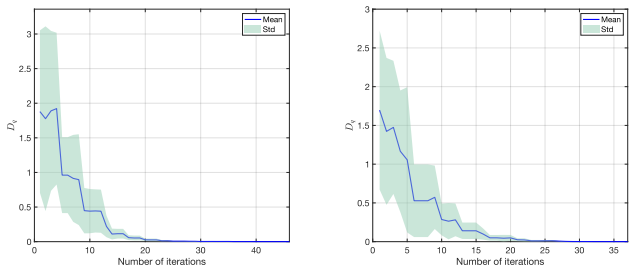
[2]This DoF advantage is known to benefit the gain improvement of BD-RIS over conventional RIS [3].



(a) $K = I = 2$, $d_s = 2$ m.

(b) $K = I = 4$, $d_s = 2$ m.

Fig. 3: WSR versus number of RIS elements for stem-connected, tree-connected (tridiagonal), and single-connected BD-RIS.



(a) $K = I = 2$ (b) $K = I = 4$

Fig. 4: Convergence behavior of the proposed algorithm for tree-connected (tridiagonal) BD-RIS.

TABLE I: Simulation Parameters

Parameter	**Value**
Number of BS antennas	$N = 16$
BS, RIS, and user heights	$h_{\mathrm{BS}} = 6\,\mathrm{m}$, $h_{\mathrm{RIS}} = 3\,\mathrm{m}$, $h_{\mathrm{u}} = 1.5\,\mathrm{m}$
BS and reference RIS location	$(0\,\mathrm{m}, 0\,\mathrm{m}), (50\,\mathrm{m}, 0\,\mathrm{m})$
User distribution	user center: $(60\,\mathrm{m}, 0\,\mathrm{m})$, radius: $10\,\mathrm{m}$
LoS path-loss parameters	$\mathrm{PL}_0 = 61.4$, $b = 2.0$, $\sigma_\zeta = 5.8$
NLoS path-loss parameters	$\mathrm{PL}_0 = 72.0$, $b = 2.92$, $\sigma_\zeta = 8.7$
Total paths per link	$L_{\mathrm{BR}} = L_{\mathrm{RU}} = 6$
BS transmit power	$P_t = 30\,\mathrm{dBm}$
Noise power	$\sigma^2 = -85\,\mathrm{dBm}$
User weight	$\omega_k = 1$, $\forall k$
Convergence threshold	$\varepsilon = \tau = 10^{-3}$
Minimum and maximum inter-panel spacing	$d_{\min} = 0.5\,\mathrm{m}$, $d_{\max} = 8\,\mathrm{m}$
Number of simplex vertices	$N_s = 5$

connected architectures show a gain of 24.53% and 23.55%, respectively. These trends confirm that optimizing d_s is particularly beneficial when the RIS panels are large and when the interconnection can be exploited by richer architectures.

Finally, Fig. 4 illustrates the convergence of the proposed algorithm for the tree-connected (tridiagonal) BD-RIS when $K = I = 2$ and $K = I = 4$. The plots show the mean and standard deviation (std) of the simplex diameter D_q (see Algorithm 1) versus the number of NM iterations. In both cases the mean value drops rapidly in the first few iterations, from about 3 m to below 0.5 m by iteration 10, and the std (shaded region) collapses to zero at a similar rate. The algorithm converges within roughly 20 iterations for both $K = I = 2$ and $K = I = 4$. The step-wise behavior is due to the shrink operation of the NM method [10], particularly as implemented in steps 27 and 34 of Algorithm 1.

V. Conclusion

In this work, we investigated the WSR maximization of a multi-panel BD-RIS-assisted outdoor-to-indoor mmWave MU-MISO system, demonstrating the significant impact of inter-panel spacing on its performance. Starting from an aggregate BD-RIS representation that accommodates arbitrary BD-RIS architectures, we formulated a WSR maximization problem subject to the BS beamforming, BD-RIS susceptance matrices and the inter-panel spacing. A two-step algorithm was then proposed that extends the pp-ADMM method to multi-panel BD-RIS and embeds the NM simplex search to tune the spacing. Simulation results demonstrated a clear trade-off between reduced channel correlation and path loss, showing that the WSR grows with spacing up to a critical point that increases with the number of RIS elements, then degrades due to increased propagation distance. By adaptively tracking this optimum, the proposed approach achieved up to approximately 25% higher WSR compared with a fixed-spacing baseline and converged within a limited number of iterations. These findings highlight the importance of geometry-aware design for BD-RIS-assisted mmWave systems.

Acknowledgment

The author thanks Z. Wu for helpful discussions and guidance on implementing pp-ADMM algorithm in [9].

References

[1] D. W. K. Ng, M. Breiling, C. Rohde, F. Burkhardt, and R. Schober, "Energy-efficient 5g outdoor-to-indoor communication: Sudas over licensed and unlicensed spectrum," *IEEE Transactions on Wireless Communications*, vol. 15, no. 5, pp. 3170–3186, 2016.

[2] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Transactions on Wireless Communications*, vol. 18, no. 11, pp. 5394–5409, 2019.

[3] S. Shen, B. Clerckx, and R. Murch, "Modeling and architecture design of reconfigurable intelligent surfaces using scattering parameter network analysis," *IEEE Transactions on Wireless Communications*, vol. 21, no. 2, pp. 1229–1243, 2022.

[4] H. Li, S. Shen, and B. Clerckx, "Beyond diagonal reconfigurable intelligent surfaces: From transmitting and reflecting modes to single-, group-, and fully-connected architectures," *IEEE Transactions on Wireless Communications*, vol. 22, no. 4, pp. 2311–2324, 2023.

[5] ——, "Beyond diagonal reconfigurable intelligent surfaces: A multi-sector mode enabling highly directional full-space wireless coverage," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 8, pp. 2446–2460, 2023.

[6] Z. Liu, X. Chu, D. López-Pérez, and N. Tang, "Deployment strategy of intelligent omni-surface-assisted outdoor-to-indoor millimeter-wave communications," *IEEE Transactions on Wireless Communications*, vol. 23, no. 12, pp. 19 172–19 182, 2024.

[7] J. Zhang and D. M. Blough, "Optimizing coverage with intelligent surfaces for indoor mmwave networks," in *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications*, 2022, pp. 830–839.

[8] R. Liang, J. Fan, H. Liu, Y. Ge, and J. Zhang, "Dual-hop hybrid irs-aided outdoor-to-indoor mmwave communications," *IEEE Communications Letters*, vol. 26, no. 12, pp. 2979–2983, 2022.

[9] Z. Wu and B. Clerckx, "Optimization of beyond diagonal ris: A universal framework applicable to arbitrary architectures," 2024. [Online]. Available: https://arxiv.org/abs/2412.15965

[10] T. Liu, "An enhanced posta based on nelder-mead simplex search and quadratic interpolation," *arXiv preprint arXiv:2405.00122*, 2024.

[11] M. Nerini, S. Shen, H. Li, M. Di Renzo, and B. Clerckx, "A universal framework for multiport network analysis of reconfigurable intelligent surfaces," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 14 575–14 590, 2024.

[12] E. N. Egashira, D. P. M. Osorio, N. T. Nguyen, and M. Juntti, "Secrecy capacity maximization for a hybrid relay-ris scheme in mmwave mimo networks," in *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, 2022, pp. 1–6.

[13] Z. Wu, M. Nerini, and B. Clerckx, "Beyond-diagonal ris architecture design and optimization under physics-consistent models," *IEEE Transactions on Wireless Communications*, vol. 25, pp. 14 086–14 100, 2026.

[14] X. Long, Y. Li, M. Pauli, B. Nuss, and T. Zwick, "Joint bs deployment and power optimization for minimum emf exposure with rl in real-world based urban scenario," *arXiv preprint arXiv:2504.05017*, 2025.

[15] Z. Wu and B. Clerckx, "Beyond-diagonal ris in multiuser mimo: Graph theoretic modeling and optimal architectures with low complexity," *IEEE Transactions on Information Theory*, pp. 1–1, 2025.